

B.E. Electrical Engineering Examination – 2022
SECOND YEAR, SECOND SEMESTER

ELECTRICAL MACHINES - II

PART - II COMPLETE SOLUTIONS

Comprehensive Academic Solution Manual

Instructions: Answer any three questions. Question no. 3 carries 18 marks. All sub-questions are solved below comprehensively for full academic coverage.

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Question 1(i)

Problem Statement:

Why a core type three phase transformer is better than other for harmonics related problems.

Thought Process

To provide an expert-level answer: 1. Identify that the core issue lies in the 3rd harmonic (triplen harmonics) components of the magnetizing current/flux. 2. Highlight that triplen harmonics are co-phaser (in phase with each other in all three phases). 3. Analyze the magnetic circuit topology: In a 3-limb core-type transformer, the sum of the fluxes must equal zero at the central junction point. Since triplen harmonic fluxes are in phase, they would have to return through the surrounding medium (air, oil, tank). 4. Conclude that the high-reluctance external path naturally suppresses triplen harmonic fluxes, preventing severe voltage distortion.

Answer:

In a three-phase system, non-linearities in the magnetic core's $B - H$ hysteresis characteristics necessitate a non-sinusoidal magnetizing current to establish a purely sinusoidal flux. The dominant distortion component is the third harmonic (and its odd multiples, known as **triplen harmonics**). Mathematically, the third harmonic currents in a balanced three-phase system can be represented as:

$$i_{a3} = I_{3m} \sin(3\omega t + \theta_3) \quad (1)$$

$$i_{b3} = I_{3m} \sin(3(\omega t - 120^\circ) + \theta_3) = I_{3m} \sin(3\omega t - 360^\circ + \theta_3) = I_{3m} \sin(3\omega t + \theta_3) \quad (2)$$

$$i_{c3} = I_{3m} \sin(3(\omega t + 120^\circ) + \theta_3) = I_{3m} \sin(3\omega t + 360^\circ + \theta_3) = I_{3m} \sin(3\omega t + \theta_3) \quad (3)$$

Since $i_{a3} = i_{b3} = i_{c3}$, triplen harmonics are completely **in phase** (co-phaser) in all three phases.

The structural superiority of a **3-phase core-type transformer** regarding harmonic problems arises directly from its magnetic circuit topology:

- Flux Return Path Constraint:** In a standard 3-limb core-type transformer, the three magnetic circuits are interconnected. By Kirchhoff's Flux Law, the sum of the fluxes entering the top yoke must equal the flux leaving it. For the fundamental component, $\phi_a + \phi_b + \phi_c = 0$, so no external return path is required. However, for the third harmonic flux, since they are in phase, they all attempt to flow in the same direction (e.g., upwards through all three limbs simultaneously).
- High Reluctance Path for Harmonics:** Because there is no fourth or fifth iron limb to provide a low-reluctance return path, the triplen harmonic fluxes ($\phi_{a3}, \phi_{b3}, \phi_{c3}$) are forced to complete their magnetic circuit through the non-magnetic paths surrounding the core, which include the transformer oil, air insulation, and the mild steel structural tank walls.
- Natural Suppression of Harmonic Flux:** The magnetic reluctance of air and transformer oil is orders of magnitude greater than that of the silicon-steel core ($R_{\text{air}} \gg R_{\text{iron}}$). Due to this exceptionally high reluctance path, the magnitude of the third harmonic flux is naturally suppressed to a very small fraction (typically less than 1% to 2%) of its unsuppressed value.
- Comparison with Other Constructions:**
 - Shell-type / 3-Phase Bank of Single-Phase Transformers:** In these configurations, each phase has a completely independent low-reluctance iron magnetic path. Consequently, the third-harmonic flux can develop fully without suppression, leading to highly distorted, peaked phase voltages that heavily stress the insulation.

Thus, the core-type three-phase transformer inherently minimizes harmonic voltage distortion and prevents dangerous insulation stresses.

Question 1(ii)

Problem Statement:

How Delta winding is used to overcome the problems of harmonics in a three phase transformers?

Thought Process

Key analytical points to address: 1. State the dual requirements of a transformer regarding flux and magnetizing current. 2. Explain how a closed Delta loop interacts with co-phaser currents. 3. Show that while triplen harmonic currents cannot flow into external lines in a delta connection, they can circulate continuously within the closed mesh loop. 4. Detail how this circulating current acts as an internal magnetomotive force (mmf) source that supplies the necessary third harmonic component of the excitation current.

Answer:

A delta (Δ) winding acts as an elegant and highly effective economic solution to harmonic problems due to its specific electrical topology. The operating principles behind this mitigation strategy are explained below:

1. **The Core Non-Linearity Dilemma:** Due to magnetic saturation in the core steel, if the excitation current is restricted to a pure sinusoid, the resulting magnetic flux waveform (ϕ) becomes distinctly flat-topped. A flat-topped flux wave contains a large third-harmonic flux component (ϕ_3), which induces highly distorted, peaked voltages in the windings. To maintain a purely sinusoidal induced voltage, the flux must remain sinusoidal, which requires a magnetizing current containing a prominent third-harmonic component (I_3).
2. **Circulating Loop Action:** In a delta-connected winding, the three phases are connected sequentially in a closed loop (mesh). As established mathematically, the triplen harmonic currents ($I_3, I_9, \dots$) in all three phases are completely in phase with each other. If we evaluate the line current (I_{line}) leaving a delta node, it is equal to the difference between two adjacent phase currents:

$$I_{\text{line}, 3} = I_{\text{phase}, a3} - I_{\text{phase}, b3} \quad (4)$$

Since $I_{\text{phase}, a3} = I_{\text{phase}, b3}$, the third-harmonic line current is identically zero. Consequently, triplen harmonic currents cannot escape into the external transmission lines. However, because they are in phase, they flow in unison around the closed mesh loop of the delta winding.

3. **Compensation Mechanism (MMF Trapping):** When a transformer is connected such that a delta winding is present, any flat-topped core flux induces a third-harmonic EMF (E_3) within the delta winding. Because the delta forms a closed path, this induced E_3 drives a circulating third-harmonic current (I_{3c}) through the low internal impedance of the winding:

$$I_{3c} = \frac{3 \cdot E_3}{3 \cdot Z_3} = \frac{E_3}{Z_3} \quad (5)$$

This circulating third-harmonic current generates an internal magnetomotive force (MMF) that directly counters the distortion. It effectively supplies the missing third-harmonic component of the magnetizing current required by the non-linear core.

4. **Restoration of Sinusoidal Conditions:** The MMF produced by this circulating current dampens the third-harmonic flux component ϕ_3 , restoring the core flux back to an almost purely sinusoidal waveform. Because the core flux is smoothed into a sinusoid, the voltages induced in the primary, secondary, and any other auxiliary windings are rendered completely sinusoidal.

Question 1(iii)

Problem Statement:

*What is the harmonics related problem with Yy connected transformer without neutral?
- Explain with proper diagram.*

Thought Process

Steps for a comprehensive response: 1. Explain that without a neutral wire, the sum of line currents must be zero ($i_a + i_b + i_c = 0$). Since third harmonic currents are equal and in-phase, their sum is $3i_{a3} = 0 \implies i_{a3} = 0$. Hence, 3rd harmonic currents are completely suppressed. 2. Discuss the consequence: The core flux becomes flat-topped due to saturation. 3. Quantify the resulting induced voltage: A flat-topped flux induces a large, in-phase 3rd-harmonic voltage in each phase, making the phase voltage waveform highly peaked. 4. Explain the phenomenon of the "Floating Neutral". 5. Provide structural and waveform diagrams showing the flattening of the flux, peaking of the voltage, and the shifting neutral point.

Answer:

A Star-Star (Y-y) connected transformer without an isolated primary neutral wire or a 4-wire return system suffer from severe operational problems due to the suppression of triplen harmonics.

1. Suppression of Third-Harmonic Current

In an ungrounded star connection, by Kirchhoff's Current Law (KCL), the sum of the line currents entering the neutral junction must be zero:

$$i_a + i_b + i_c = 0 \quad (6)$$

For triplen harmonics, we know that $i_{a3} = i_{b3} = i_{c3}$. Substituting this into the KCL equation yields:

$$i_{a3} + i_{b3} + i_{c3} = 3i_{a3} = 0 \implies i_{a3} = i_{b3} = i_{c3} = 0 \quad (7)$$

Thus, the lack of a primary neutral wire completely suppresses the flow of the third-harmonic component of the magnetizing current.

2. Flattening of Flux and Peaking of Phase Voltage

Because the non-linear iron core is denied the necessary third-harmonic component of current, the magnetic flux (ϕ) cannot remain sinusoidal; instead, it becomes severely **flat-topped** due to core saturation at higher excitation levels.

By Fourier analysis, a flat-topped flux wave contains a large, in-phase third-harmonic flux component ($\phi_3 = \Phi_{3m} \sin(3\omega t)$). The induced electromotive force (EMF) in any phase is given by Faraday's Law:

$$e_{\text{phase}} = -\frac{d\phi}{dt} = -\frac{d}{dt} [\Phi_{1m} \sin(\omega t) + \Phi_{3m} \sin(3\omega t)] \quad (8)$$

$$= -\omega\Phi_{1m} \cos(\omega t) - 3\omega\Phi_{3m} \cos(3\omega t) \quad (9)$$

Notice the multiplier **3** ahead of the third harmonic term. Even a relatively small third-harmonic flux component (e.g., 10%) yields a very large third-harmonic voltage component ($3 \times 10\% = 30\%$). Because this harmonic voltage is added in opposition to the fundamental at the peak of the wave, it distorts the phase voltage into an extremely sharp, **highly peaked waveform**. The peak value can soar to 140% – 150% of normal nominal levels, threatening insulation integrity.

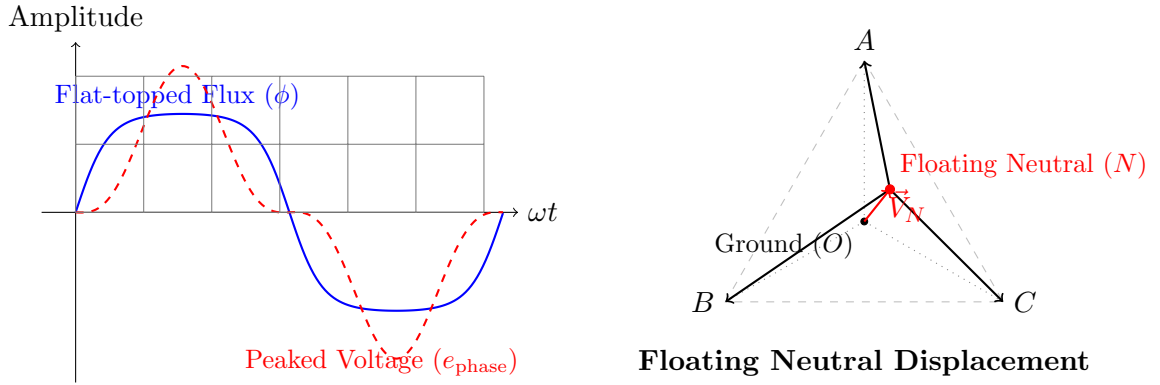
3. The Floating Neutral Phenomenon

Since the third-harmonic voltages in all three phases are equal and in phase ($e_{a3} = e_{b3} = e_{c3} = e_3$), they do not appear across the external lines:

$$v_{ab3} = e_{a3} - e_{b3} = 0 \quad (10)$$

Line-to-line voltages remain relatively balanced. However, the third-harmonic voltage appears fully between each line and the physical neutral point. This forces the neutral point potential (V_N) to oscillate violently at a frequency of 3ω (typically 150 Hz) relative to ground. This instability is known as a **Floating Neutral**.

4. Illustrative Waveforms and Schematic Diagrams



Question 2(i)

Problem Statement:

Draw the connection diagram and phasor diagram of the following connections: a) *Yy6*, b) *Dz6* and c) *Dy1*

Thought Process

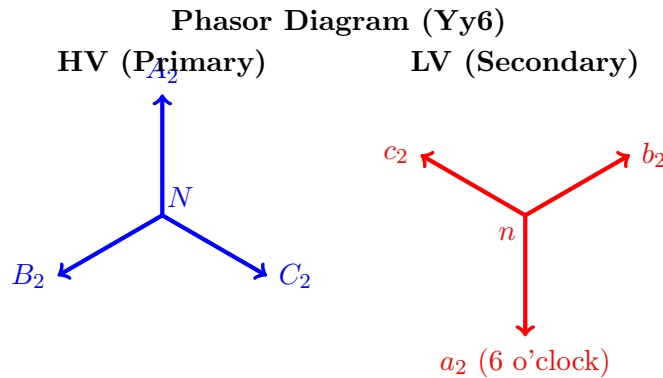
The question asks for the connection and phasor diagrams for three standard three-phase transformer configurations using the clock convention: 1. ****Yy6****: Star-Star with a 180° phase displacement (6 o'clock). 2. ****Dz6****: Delta-Zigzag star with a 180° phase displacement (6 o'clock). 3. ****Dy1****: Delta-Star with a -30° phase displacement (1 o'clock).

Answer:

The standard clock convention specifies that the High Voltage (HV) phase voltage vector acts as the reference minute hand fixed at the 12 o'clock position. The Low Voltage (LV) terminal positions act as the hour hand.

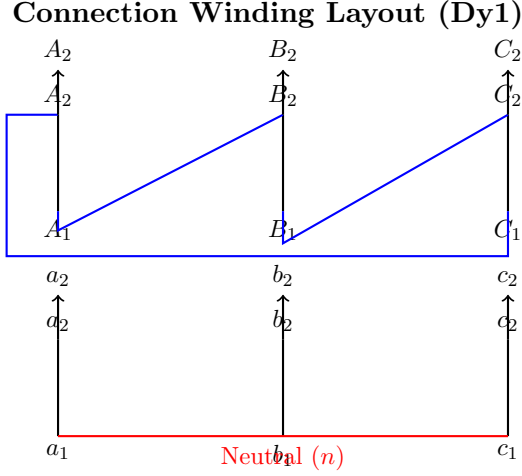
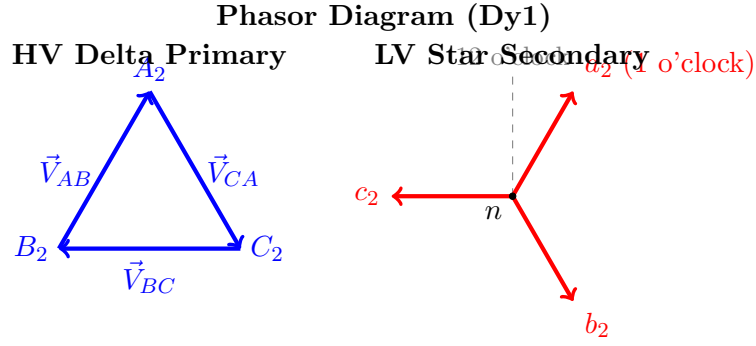
Configuration 1: Yy6 (Star-Star, 180° Phase Displacement)

In a Yy6 connection, the primary and secondary are both star-connected, but the secondary windings are cross-reversed relative to the primary terminals to yield a 180° phase inversion (6 o'clock position).



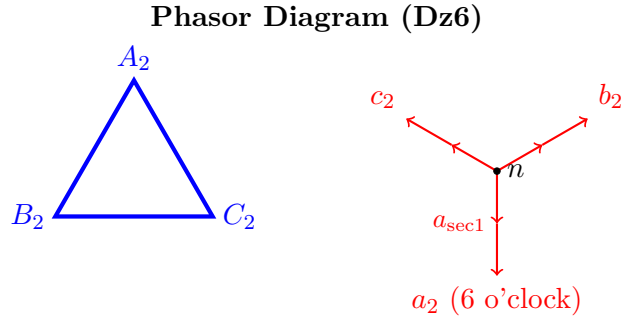
Configuration 2: Dy1 (Delta-Star, -30° Shift, 1 o'clock)

In a Dy1 transformer, the high-voltage primary winding is wired in delta (Δ), and the low-voltage secondary is arranged in a conventional star configuration, yielding a 30° lagging phase angle.



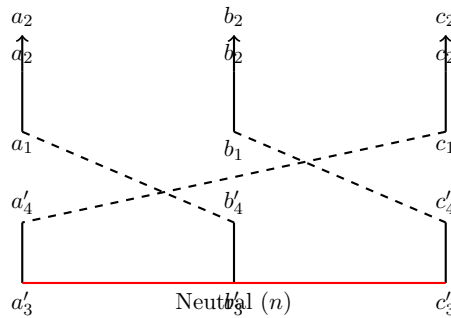
Configuration 3: Dz6 (Delta-Zigzag Star, 180° Displacement, 6 o'clock)

In a Dz6 configuration, the primary winding is wired in delta (Δ). The secondary winding is split into two equal sections on separate limbs and cross-connected in a zigzag profile to cancel zero-sequence harmonics and provide a stable 180-degree phase shift.



Connection Winding Layout (Dz6 Secondary Mesh)

Secondary Side Interconnected Zigzag Winding (6 o'clock)



Question 2(ii)

Problem Statement:

What will happen if a Dy1 connected transformer is paralleled with a Yd11 connected transformer? Show how they can be connected properly with connection modification.

Thought Process

Step-by-step resolution: 1. Determine the net phase angular difference between their secondary terminals when excited by a common primary bus: $+30^\circ - (-30^\circ) = 60^\circ$. 2. Calculate the quantitative consequences of direct paralleling: Compute the resulting voltage difference ($\Delta V = 2V_2 \sin(60^\circ/2) = V_2$). Explain that this unmitigated phase difference acts as a direct short-circuit path. 3. Propose the necessary physical modifications: Show that by altering the external phase sequences—specifically interchanging two primary input supply lines and cross-swapping the corresponding secondary lines—the Yd11 transformer can be converted into a Yd1 equivalent configuration.

Answer:

1. Quantitative Evaluation of Direct Paralleling

To operate two transformer banks safely in parallel, their secondary output voltages must satisfy four critical conditions: identical voltage magnitude ratios, identical frequency, identical phase sequence, and **zero relative phase displacement**.

Let us evaluate the phase parameters of the two given configurations:

- **Transformer 1 (Dy1):** Belongs to Phase Group 3, introducing a fixed phase displacement of **-30°** (the secondary line voltage lags the primary line voltage by 30°).
- **Transformer 2 (Yd11):** Belongs to Phase Group 4, introducing a fixed phase displacement of **$+30^\circ$** (the secondary line voltage leads the primary line voltage by 30°).

If both transformers are tied directly to the same primary network, the net angular phase divergence (α) between their corresponding secondary phase terminals (a_{Dy1} and a_{Yd11}) is:

$$\alpha = +30^\circ - (-30^\circ) = 60^\circ \quad (11)$$

Assuming both units have equal secondary line voltage magnitudes of V_2 , the structural voltage difference ($\Delta \vec{V}$) developed across the paralleling contact points is computed via phasor subtraction:

$$|\Delta \vec{V}| = 2 \cdot V_2 \cdot \sin\left(\frac{\alpha}{2}\right) = 2 \cdot V_2 \cdot \sin(30^\circ) = 2 \cdot V_2 \cdot (0.5) = V_2 \quad (12)$$

The structural voltage mismatch is exactly equal to the full nominal secondary line voltage. Because the internal leakage impedance (Z_{e1}, Z_{e2}) of a distribution transformer is very small, this voltage difference induces an massive circulating fault current (I_c):

$$I_c = \frac{|\Delta \vec{V}|}{Z_{e1} + Z_{e2}} = \frac{V_2}{Z_{e1} + Z_{e2}} \quad (13)$$

This circulating current will act as a severe line-to-line short circuit. This will result in **immediate catastrophic structural burn-out of the windings**. **Direct paralleling is strictly prohibited.**

2. Corrective Connection Modification Strategy

While transformers from Group 3 and Group 4 cannot normally be paralleled, we can resolve the 60° phase discrepancy by systematically interchanging the external phase connections on both the primary input and secondary output terminals of the **Yd11** transformer. This modification alters its effective phase shift from $+30^\circ$ to -30° .

Phase Sequence Reversal Modification Steps:

1. **Primary Side Modification:** Disconnect lines B and C from the primary terminal entries of the Yd11 transformer and swap them. Terminal A_2 remains connected to Phase A . Connect Phase B to terminal C_2 , and connect Phase C to terminal B_2 . This forces a primary phase sequence reversal from $A - B - C$ to $A - C - B$.
2. **Resulting Phase Shift Inversion:** Reversing the primary phase sequence shifts the internal phasor rotation. As a result, the $+30^\circ$ phase lead of the Yd11 transformer becomes a -30° phase lag relative to the original system frame. This effectively converts the Yd11 transformer into a Yd1 operating configuration.
3. **Secondary Side Correction:** Because the internal phase sequence was reversed, the secondary line terminals now rotate in a reversed sequence ($a - c - b$). To restore the correct standard phase rotation ($a - b - c$) before tying into the common load busbar, interchange secondary output cables b and c . Connect terminal a_2 to bus Phase a , terminal b_2 to bus Phase c , and terminal c_2 to bus Phase b .

Following this dual-swap modification, the secondary output voltages of both transformers are brought perfectly in phase ($\Delta \vec{V} = 0$), enabling safe, stable parallel operation.

Question 3(i)

Problem Statement:

If two identical single-phase transformers have tapings at 50% and 86.7% then show that they can be connected in Scott connection to convert a three to a two phase system.

Thought Process

Analytical derivation plan: 1. Identify the two transformers: Main Transformer and Teaser Transformer. 2. Define the primary turn structure: Let the Main transformer primary have N_P turns, with a center tap at 50% ($N_P/2$). Let the Teaser transformer primary utilize a $\frac{\sqrt{3}}{2} \approx 86.7\%$ tap. 3. Apply standard balanced 3-phase line voltage phasors ($\vec{V}_{AB}, \vec{V}_{BC}, \vec{V}_{CA}$) and use geometric phasor analysis to show that the voltage across the teaser primary ($\vec{V}_{AD}$) is perpendicular to the main primary voltage ($\vec{V}_{BC}$) and has a magnitude of $\frac{\sqrt{3}}{2}V_L$. 4. Calculate the secondary output induced voltages for both units using their respective turns ratios, proving that $V_{S1} = V_{S2}$ in magnitude and orthogonal.

Answer:

The Scott connection utilizes two distinct single-phase transformers—the **Main Transformer** and the **Teaser Transformer**—to convert a balanced three-phase system into a balanced two-phase system (or vice versa).

1. Winding Topology Parameters

Let both identical single-phase transformers have an available primary total turn count of N_P and a secondary turn count of N_S .

- **Main Transformer Winding:** The primary winding is connected directly between 3-phase supply lines B and C . It features a precise center-tap at the 50% mark, designated as terminal D . The number of turns between B and D is $N_P/2$, and between D and C is $N_P/2$.
- **Teaser Transformer Winding:** The primary winding is connected between 3-phase line terminal A and the main primary center-tap D . It utilizes an 86.7% tap ($\frac{\sqrt{3}}{2}N_P$ turns).

2. Phasor Derivation and Voltage Analysis

Let the balanced three-phase input line-to-line voltage magnitude be V_L . Taking the line voltage $\vec{V}_{BC}$ as our reference phasor along the horizontal axis:

$$\vec{V}_{BC} = V_L \angle 0^\circ = V_L + j0 \quad (14)$$

The remaining symmetrical balanced line voltage phasors are:

$$\vec{V}_{AB} = V_L \angle 120^\circ = V_L \left(-0.5 + j\frac{\sqrt{3}}{2} \right) \quad (15)$$

$$\vec{V}_{CA} = V_L \angle -120^\circ = V_L \left(-0.5 - j\frac{\sqrt{3}}{2} \right) \quad (16)$$

Since terminal D is the exact geometric midpoint of the main primary winding across lines B and C , the voltage from B to D is:

$$\vec{V}_{BD} = \frac{1}{2}\vec{V}_{BC} = \frac{1}{2}V_L \angle 0^\circ = 0.5V_L \quad (17)$$

Now, let us determine the voltage across the teaser transformer primary winding, which is connected between terminal A and node D , using loop phasor addition:

$$\vec{V}_{AD} = \vec{V}_{AB} + \vec{V}_{BD} \quad (18)$$

$$\vec{V}_{AD} = V_L \left(-0.5 + j\frac{\sqrt{3}}{2} \right) + 0.5V_L \quad (19)$$

$$\vec{V}_{AD} = -0.5V_L + j\frac{\sqrt{3}}{2}V_L + 0.5V_L = j\frac{\sqrt{3}}{2}V_L \quad (20)$$

Converting this rectangular result to polar form yields:

$$\vec{V}_{AD} = \frac{\sqrt{3}}{2}V_L \angle 90^\circ \approx 0.8667V_L \angle 90^\circ \quad (21)$$

This confirms two critical conditions: 1. The voltage magnitude applied to the teaser primary is exactly $\frac{\sqrt{3}}{2}$ (86.7%) of the line voltage V_L . 2. The teaser voltage phasor $\vec{V}_{AD}$ leads the main transformer voltage phasor $\vec{V}_{BC}$ by exactly 90° in time quadrature.

3. Secondary Turn Balance Verification

Now, let us check the secondary output voltages ($V_{S,\text{main}}$ and $V_{S,\text{teaser}}$) induced across the secondary windings, both of which have N_S turns:

- **Main Transformer Secondary Output:**

$$V_{S,\text{main}} = V_{BC} \cdot \left(\frac{N_S}{N_P} \right) = V_L \cdot \left(\frac{N_S}{N_P} \right) \quad (22)$$

- **Teaser Transformer Secondary Output:** Knowing that its active primary turns are $N_{PT} = \frac{\sqrt{3}}{2}N_P$:

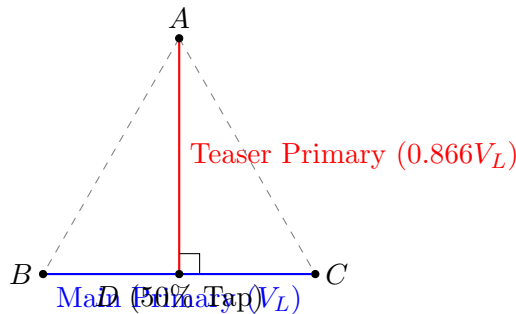
$$V_{S,\text{teaser}} = V_{AD} \cdot \left(\frac{N_S}{N_{PT}} \right) = \left(\frac{\sqrt{3}}{2}V_L \right) \cdot \frac{N_S}{\left(\frac{\sqrt{3}}{2}N_P \right)} = V_L \cdot \left(\frac{N_S}{N_P} \right) \quad (23)$$

Comparing the two outputs:

$$|V_{S,\text{main}}| = |V_{S,\text{teaser}}| = V_L \left(\frac{N_S}{N_P} \right) \quad (24)$$

Because the input primary voltages $\vec{V}_{AD}$ and $\vec{V}_{BC}$ are orthogonal (90° displacement), the corresponding induced secondary output voltages $\vec{V}_{S,\text{main}}$ and $\vec{V}_{S,\text{teaser}}$ are also equal in magnitude and displaced by exactly 90° . This satisfies the requirements for a balanced two-phase electrical system.

Scott Connection Primary Voltage Phasor Geometry



Question 3(ii)

Problem Statement:

In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 30 kW at 220 V and main transformer supplies 0.85 power factor lagging load of 50 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above.

Thought Process

Given parameters: - Primary line voltage $V_L = 3300$ V. Secondary voltage $V_S = 220$ V. - Turn transformation ratio base: $v = N_P/N_S = 3300/220 = 15$. - Teaser Winding Load: $P_T = 30$ kW, $\cos \phi_T = 0.75$ leading. - Main Winding Load: $P_M = 50$ kW, $\cos \phi_M = 0.85$ lagging.

Step-by-step mathematical plan: 1. Establish standard coordinate phasor vectors for voltages: $\vec{V}_{BC} = 3300\angle 0^\circ$, meaning secondary main voltage $\vec{V}_{M2} = 220\angle 0^\circ$. Teaser voltage $\vec{V}_{AD} = 2857.88\angle 90^\circ$, meaning secondary teaser voltage $\vec{V}_{T2} = 220\angle 90^\circ$. 2. Compute secondary load currents as complex vectors. 3. Map secondary currents back to the primary inputs using MMF balance relationships. 4. Perform complex vector operations to find polar magnitudes and angles for $\vec{I}_A, \vec{I}_B, \vec{I}_C$.

Answer:

1. Calculations for the Secondary Load Currents

Let us define the secondary voltage references based on our primary orientation:

- Secondary Main Winding Voltage: $\vec{V}_{M2} = 220\angle 0^\circ$ V
- Secondary Teaser Winding Voltage: $\vec{V}_{T2} = 220\angle 90^\circ$ V

We calculate the secondary current magnitudes from the specified power delivery constraints:

- **Teaser Secondary Current (I_{T2}):** Given $P_T = 30,000$ W and $\cos \phi_T = 0.75$ leading ($\phi_T = +\arccos(0.75) = +41.41^\circ$):

$$I_{T2} = \frac{P_T}{V_{T2} \cdot \cos \phi_T} = \frac{30000}{220 \cdot 0.75} = \frac{30000}{165} = 181.818 \text{ A} \quad (25)$$

Expressed as a phasor relative to the reference frame ($\vec{V}_{T2}$ is at 90°):

$$\vec{I}_{T2} = 181.818\angle(90^\circ + 41.41^\circ) = 181.818\angle 131.41^\circ \text{ A} \quad (26)$$

- **Main Secondary Current (I_{M2}):** Given $P_M = 50,000$ W and $\cos \phi_M = 0.85$ lagging ($\phi_M = -\arccos(0.85) = -31.79^\circ$):

$$I_{M2} = \frac{P_M}{V_{M2} \cdot \cos \phi_M} = \frac{50000}{220 \cdot 0.85} = \frac{50000}{187} = 267.380 \text{ A} \quad (27)$$

Expressed as a phasor relative to the reference frame ($\vec{V}_{M2}$ is at 0°):

$$\vec{I}_{M2} = 267.380\angle -31.79^\circ \text{ A} \quad (28)$$

2. Derivation of Primary Line Currents

The nominal base turns ratio for the main transformer is:

$$v = \frac{N_P}{N_S} = \frac{3300}{220} = 15 \quad (29)$$

Step A: Primary Line Current $\vec{I}_A$

Line current $\vec{I}_A$ feeds the primary winding of the teaser transformer. By MMF balance:

$$\vec{I}_A \cdot \left(\frac{\sqrt{3}}{2} N_P \right) = \vec{I}_{T2} \cdot N_S \implies \vec{I}_A = \frac{2}{\sqrt{3}} \cdot \frac{1}{v} \cdot \vec{I}_{T2} \quad (30)$$

Substituting the values:

$$\vec{I}_A = \frac{2}{\sqrt{3}} \cdot \frac{1}{15} \cdot 181.818 \angle 131.41^\circ = 14.001 \angle 131.41^\circ \text{ A} \quad (31)$$

Converting $\vec{I}_A$ to rectangular coordinates:

$$\vec{I}_A = 14.001 \cdot [\cos(131.41^\circ) + j \sin(131.41^\circ)] \quad (32)$$

$$= 14.001 \cdot [-0.6614 + j0.7500] = -9.260 + j10.501 \text{ A} \quad (33)$$

Step B: Primary Line Currents $\vec{I}_B$ and $\vec{I}_C$

The primary line currents entering terminals B and C consist of two components: the main primary load current ($\vec{I}_{M1}$) flowing from B to C , and the teaser primary current ($\vec{I}_A$) entering the center tap D and splitting equally in opposite directions.

$$\vec{I}_{M1} = \frac{\vec{I}_{M2}}{v} = \frac{267.380 \angle -31.79^\circ}{15} = 17.825 \angle -31.79^\circ \text{ A} \quad (34)$$

Converting $\vec{I}_{M1}$ to rectangular form:

$$\vec{I}_{M1} = 17.825 \cdot [\cos(-31.79^\circ) + j \sin(-31.79^\circ)] \quad (35)$$

$$= 17.825 \cdot [0.8500 - j0.5268] = 15.151 - j9.390 \text{ A} \quad (36)$$

Now, applying current component distribution rules:

$$\vec{I}_B = \vec{I}_{M1} - \frac{\vec{I}_A}{2} \quad \text{and} \quad \vec{I}_C = -\vec{I}_{M1} - \frac{\vec{I}_A}{2} \quad (37)$$

Let us calculate the shared center-tap return component:

$$\frac{\vec{I}_A}{2} = \frac{-9.260 + j10.501}{2} = -4.630 + j5.251 \text{ A} \quad (38)$$

Calculating $\vec{I}_B$:

$$\vec{I}_B = (15.151 - j9.390) - (-4.630 + j5.251) \quad (39)$$

$$= 19.781 - j14.641 \text{ A} \quad (40)$$

Converting to polar coordinates:

$$|I_B| = \sqrt{(19.781)^2 + (-14.641)^2} = \sqrt{391.288 + 214.359} = 24.610 \text{ A} \quad (41)$$

$$\theta_B = \arctan\left(\frac{-14.641}{19.781}\right) = -36.51^\circ \quad (42)$$

$$\vec{I}_B = 24.610 \angle -36.51^\circ \text{ A} \quad (43)$$

Calculating $\vec{I}_C$:

$$\vec{I}_C = -(15.151 - j9.390) - (-4.630 + j5.251) \quad (44)$$

$$= -10.521 + j4.139 \text{ A} \quad (45)$$

Converting to polar coordinates:

$$|I_C| = \sqrt{(-10.521)^2 + (4.139)^2} = \sqrt{110.691 + 17.131} = 11.306 \text{ A} \quad (46)$$

$$\theta_C = 180^\circ - \arctan\left(\frac{4.139}{10.521}\right) = 180^\circ - 21.48^\circ = 158.52^\circ \quad (47)$$

$$\vec{I}_C = 11.306 \angle 158.52^\circ \text{ A} \quad (48)$$

3. Validation Check

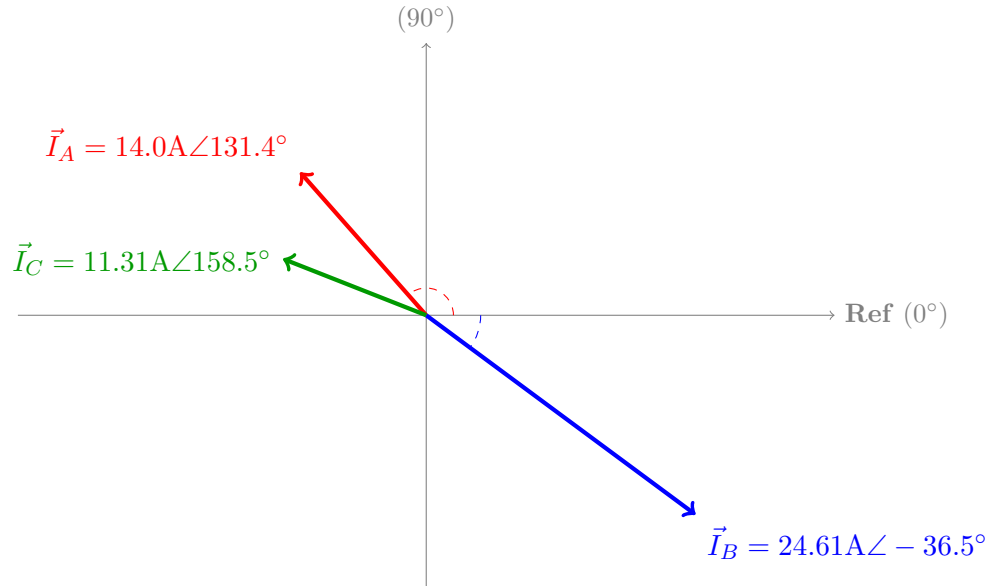
Summing the three computed line current vectors confirms compliance with Kirchhoff's Current Law:

$$\vec{I}_A + \vec{I}_B + \vec{I}_C = (-9.260 - 10.521 + 19.781) + j(10.501 + 4.139 - 14.640) \quad (49)$$

$$= 0 + j0 \quad (50)$$

4. Primary Line Current Phasor Diagram Plot

The calculated current phasors are plotted below with respect to the common reference frame:



Question 4(i)

Problem Statement:

Taps should be provided at the middle of the high voltage winding- explain

Thought Process

Key engineering arguments to present: 1. Explain the selection of the HV side: Lower operational currents mean reduced contact arcing, and a higher number of turns allows for finer step resolution. 2. Explain the centralized middle location: Placing taps at the center maintains magnetic symmetry, minimizing dangerous mechanical axial forces during external short-circuit events. It also protects the tap-changer contacts from severe impulse voltage surges.

Answer:

In power and distribution transformers, voltage regulation taps are almost universally located on the High Voltage (HV) winding rather than the Low Voltage (LV) winding, and are placed specifically at the **geometric middle** of the winding string. This design choice is driven by both electrical and mechanical engineering considerations:

1. Why Taps are Placed on the High Voltage (HV) Winding

- **Arc Suppression and Current Mitigation:** The HV winding carries a significantly lower current than the high-current LV winding ($I_{HV} \ll I_{LV}$). Placing the tap changer on the HV side minimizes the current that the physical contact switches must interrupt, reducing arcing, contact wear, and the required physical size of the switch mechanism.
- **Fine Tuning Resolution:** The HV winding has a much larger total number of turns (N) than the LV winding. This large turn count allows for precise fractional turn adjustments (e.g., standard $\pm 1.25\%$ or $\pm 2.5\%$ regulation steps), which would be physically impossible on an LV winding with few turns.
- **Physical Accessibility:** In standard concentric transformer construction, the HV winding is wound on the outside of the core structure, over the inner LV winding. This makes it physically easier to route tap leads out to the tap-changer compartment.

2. Why Taps are Placed specifically at the Geometric Middle

- **Preservation of Magnetic Symmetry (Axial Force Reduction):** During an external short-circuit fault, transformers experience massive electrodynamic forces. If turns are cut out from one extreme end of a winding, a significant vertical magnetic asymmetry is created between the primary and secondary windings. This asymmetry generates large **axial mechanical forces** that can compress, distort, or rip the windings from their structural supports. Splitting the winding and placing the taps at the geometric center ensures that turns are removed symmetrically from both the top and bottom halves, maintaining magnetic balance.
- **Impulse Voltage Surge Protection:** When a high-frequency lightning or switching impulse surge hits a transformer terminal, the initial voltage distribution along the winding is non-linear and is governed entirely by the winding's capacitance network. This creates an extremely steep voltage gradient across the line-end turns. The center of the winding is shielded from these initial peak dielectric stresses.

Question 4(ii)

Problem Statement:

Why either resistor or reactor is used in an On-load tap changer(OLTC)?

Thought Process

Core analytical principle: An OLTC must switch from one tap to an adjacent tap without breaking the load circuit (which would cause massive arcing and power interruption) and without short-circuiting the turns between adjacent taps (which would cause a destructive circulating current). The transition resistor or reactor acts as a temporary current-limiting impedance during the switching sequence.

Answer:

An On-Load Tap Changer (OLTC) must alter the active turn ratio of a transformer without interrupting the continuous power flow to the load, and without creating destructive internal fault conditions. This dual requirement creates an engineering challenge during the transition from an active tap (e.g., Tap 1) to an adjacent tap (e.g., Tap 2).

1. **The Open-Circuit Hazard:** If the selector switch disconnected completely from Tap 1 before making contact with Tap 2, the primary circuit would be momentarily opened. This open circuit would cause a massive, destructive electric arc across the opening contacts and interrupt power to the load.
2. **The Short-Circuit Hazard:** To prevent an open circuit, the switch must make contact with Tap 2 before breaking contact with Tap 1. However, during this overlapping interval, Tap 1 and Tap 2 are briefly tied together. Because there is an inherent voltage difference (ΔV_{step}) between adjacent taps, bridging them directly creates a closed loop with very low resistance. This would drive a massive local **circulating current** ($I_{\text{circ}} = \Delta V_{\text{step}} / Z_{\text{internal}}$) through the shorted turns, quickly overheating and destroying the contacts and windings.

To resolve this problem, a temporary current-limiting transition impedance—either a **resistor** or a **reactor**—is introduced during the switching interval.

- **Function of the Impedance:** During the brief transitional moment when both taps are bridged, the resistor or reactor is connected in series within the local loop. This limits the circulating current to a safe value, while providing a continuous path for the load current to flow without interruption.

Question 4(iii)

Problem Statement:

Describe the operation of a resistor type tap changers.

Thought Process

Plan for a high-scoring response: 1. Explain that resistor-type OLTCs rely on a high-speed diverter switch combined with two transition resistors (R_1, R_2) to quickly perform the transition (typically within milliseconds via spring action). 2. Outline the step-by-step sequence of operations as the unit shifts from Tap 1 to Tap 2. 3. Draw a clear schematic circuit diagram showing the layout of the tap selectors, transition resistors, and diverter contacts.

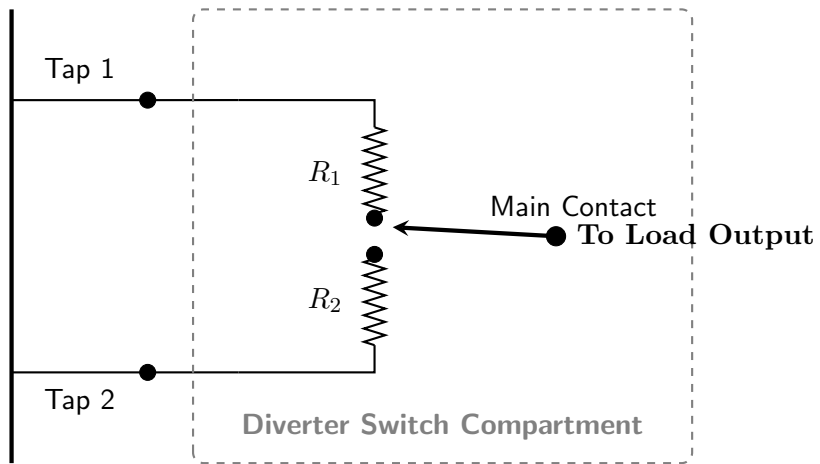
Answer:

Resistor-type On-Load Tap Changers utilize a high-speed spring-loaded **diverter switch** mechanism and temporary transition resistors to move smoothly between tap positions. The entire switching sequence occurs within milliseconds, minimizing the duration of resistor heating and contact arcing.

Step-by-Step Operational Sequence

Let us consider transitioning the connection from **Tap 1** to **Tap 2**:

- 1. Normal Steady State (Position 1):** The selector switch is physically touching Tap 1. The diverter switch is closed on its main contact side, allowing the load current (I_L) to flow directly from Tap 1 to the external output neutral line without passing through any resistors.
- 2. Pre-selection:** While power flows normally through Tap 1, the adjacent selector branch moves to pre-select **Tap 2**. No current flows through this new path yet because the corresponding diverter contact remains open.
- 3. First Transition Step:** The high-speed spring mechanism fires, rapidly moving the diverter contact. It opens the direct main path from Tap 1 and instantly introduces transition resistor **R_1** in series with Tap 1. The load current now flows entirely through R_1 .
- 4. Bridging Position (Dual Resistor State):** As the diverter continues its rapid sweep, it closes the contact to Tap 2 through transition resistor **R_2** , while maintaining the path through R_1 . At this instant, both Tap 1 and Tap 2 are connected to the output simultaneously. The local circulating current between the taps is safely limited by the sum of the two resistors ($R_1 + R_2$), and the load current is shared between the two branches.
- 5. Third Transition Step:** The diverter continues moving and opens the contact to Tap 1, breaking the circulating loop. The load current is now supplied solely from Tap 2, flowing in series through resistor R_2 .
- 6. Final Steady State (Position 2):** The diverter completes its travel and closes its main contact on the Tap 2 side, shorting out resistor R_2 . The load current now flows directly from Tap 2 to the output line, completing the tap-changing operation.



Question 5(i)

Problem Statement:

Write Short notes on: Full wave and chopped wave impulse voltage tests on a transformer for detection of faults in the winding.

Thought Process

Key details to include: 1. Explain the purpose of impulse testing: Simulating high-voltage transients from lightning or switching surges using a standard wave shape (1.2/50 μs). 2. Detail the Full-Wave Test: Application of the unchopped surge wave, and fault detection via comparing voltage and neutral-current oscillograms against baseline calibration shots. 3. Detail the Chopped-Wave Test: Using a rod gap to suddenly cut the voltage wave to zero after 2–6 μs . This simulates a line flashover and creates an extremely steep falling edge (dv/dt).

Answer:

During normal operation, power transformers are exposed to high-voltage transient surges caused by lightning strikes and network switching operations. To verify that the transformer's insulation system can withstand these dielectric stresses, units are subjected to high-voltage impulse tests using an external impulse generator. The standard unchopped testing waveform has a front time of 1.2 μs and a tail time of 50 μs (referred to as a 1.2/50 μs wave).

1. Full Wave Impulse Voltage Test

The full wave test applies the complete, unchopped 1.2/50 μs surge wave at a specified Basic Insulation Level (BIL) voltage directly to the transformer line terminals.

- **Fault Detection Mechanism:** Highly sensitive high-speed Cathode Ray Oscilloscopes (CROs) record the transient voltage wave shape applied across the terminal and the resulting current waveform flowing out of the grounded neutral point.
- **Analysis Method:** Before applying the full test voltage, a low-voltage calibration shot (typically at 50% to 70% BIL) is applied to establish a baseline waveform signature. If an insulation failure occurs during the full-scale test (such as a breakdown between adjacent turns or coils), the sudden change in internal impedance alters the wave profile. ****Any deviation, high-frequency ringing, or sudden drop in the neutral current oscillogram compared to the baseline signature indicates an internal winding insulation fault.****

2. Chopped Wave Impulse Voltage Test

In actual operation, a traveling lightning surge can cause a flashover across an adjacent line insulator string or surge arrester gap. This sudden flashover instantly grounds the line, causing the voltage to collapse to zero almost instantaneously. This condition is simulated using the Chopped Wave Test.

- **Test Setup:** A controlled spark gap (such as a rod or sphere gap) is connected in parallel with the transformer terminal. The gap is configured to flash over and ground the surge wave after a brief delay, typically between ****2 μs and 6 μs **** from the start of the impulse.
- **Severe Voltage Stress Gradient:** The sudden flashover creates an extremely steep falling edge with a massive rate of voltage collapse (high negative dv/dt). This rapid voltage drop generates extreme dielectric stress across the inter-turn insulation of the coils closest to the line terminal.

- **Fault Detection Mechanism:** Because the chopped wave's rapid collapse causes high-frequency oscillations that complicate real-time waveform analysis, faults are typically identified by applying a final full-wave impulse immediately after the chopped wave test. If the neutral current oscillogram from this final full wave matches the initial baseline full-wave signature perfectly, the insulation is verified to be intact. Any discrepancy reveals that the steep chopped wave caused an inter-turn insulation failure.

Question 5(ii)

Problem Statement:

Write Short notes on: Development of voltage stress along the winding of a three phase transformer for input impulse and RMS voltage.

Thought Process

Key physical concepts to contrast: 1. Power Frequency (RMS) Conditions: The circuit is dominated by winding resistance and inductance; the capacitive path is an open circuit. This results in a completely linear voltage distribution and uniform voltage stress across all turns. 2. Impulse Voltage Conditions: The extremely steep surge front acts as a high-frequency input, rendering the inductive paths temporary open circuits. The voltage distribution is governed entirely by the winding's capacitance network. This produces a highly non-linear exponential initial voltage drop, concentrating up to 70 – 80% of the total surge voltage across the first 10 – 15% of the turns nearest the line terminal.

Answer:

The voltage distribution and resulting dielectric stress along the length of a transformer winding depend heavily on the frequency of the applied voltage. The behavior under normal power frequency (RMS) conditions contrasts sharply with the behavior under high-voltage transients (Impulse surges).

1. Voltage Stress Under Normal Power Frequency (RMS) Voltage

At standard power operating frequencies (50 Hz), the capacitive reactance of the winding ($X_C = 1/\omega C$) is extremely high and acts as an open circuit, allowing capacitive leakage currents to be neglected. The voltage distribution along the winding is governed entirely by the series resistance (R) and leakage inductance (L) of the turns.

Because these parameters are uniformly distributed along the length of a standard winding, the steady-state voltage decreases ****completely linearly**** from the high-voltage line terminal (100% voltage) down to the grounded neutral terminal (0% voltage). As a result, the voltage gradient ($\frac{dv}{dx}$) is constant throughout the winding. Each turn experiences the same low, uniform dielectric stress.

2. Voltage Stress Under Transient Impulse Voltage

An impulse surge features an exceptionally steep front that acts as an extremely high-frequency transient (in the megahertz range). At these high frequencies, the inductive reactance ($X_L = \omega L$) becomes extremely large and acts as a temporary open circuit. Consequently, the initial voltage distribution is governed entirely by the winding's complex capacitance network:

- **Series Capacitance (C_s):** The inter-turn and inter-coil capacitances along the winding string.
- **Shunt Capacitance (C_g):** The capacitance between the winding coils and the grounded iron core or structural tank wall.

When the surge hits the line terminal, current must charge the shunt capacitances C_g to ground as it travels down the winding through the series capacitances C_s . Because current continuously leaks to ground through C_g , the current flowing through C_s decreases exponentially as it moves farther from the line terminal.

The initial voltage distribution along the winding follows an exponential decay curve governed by the distribution factor α :

$$V(x) = V_0 \cdot \frac{\sinh\left(\alpha \cdot \frac{x}{l}\right)}{\sinh(\alpha)} \quad \text{where} \quad \alpha = \sqrt{\frac{C_g}{C_s}} \quad (51)$$

Here, x is the distance from the grounded neutral and l is the total winding length. In typical power transformers, α ranges from 5 to 30.

This high α value causes a highly non-linear initial voltage distribution. Up to **70% to 80% of the total impulse voltage drop occurs across the first 10% to 15% of the turns** closest to the line terminal. This creates an extremely high localized voltage gradient and severe dielectric stress on the insulation of the end turns, requiring reinforced insulation or electrostatic shielding in those areas.

